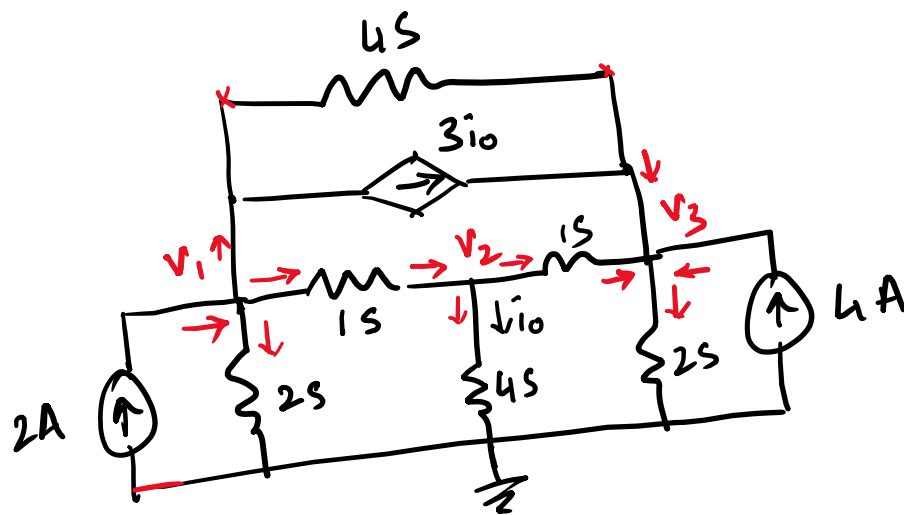




$$i_0 = v_2(4)$$

$$\text{node } v_1: 2 = v_1(2) + (v_1 - v_2)(1) + 3i_0 + (v_1 - v_3)(4)$$

$$\text{node } v_2: (v_1 - v_2)(1) = v_2(4) + (v_2 - v_3)(1)$$

$$\text{node } v_3: 4 + (v_2 - v_3)(1) + (v_1 - v_3)(4) + 3i_0 = v_3(2)$$

$$2 = 7v_1 + 11v_2 - 4v_3$$

$$0 = -v_1 + 6v_2 - v_3$$

$$-4 = 4v_1 + 13v_2 - 7v_3$$

$$A X = B$$

$$\begin{bmatrix} 7 & 11 & -4 \\ -1 & 6 & -1 \\ 4 & 13 & -7 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ 0 \\ -4 \end{bmatrix}$$

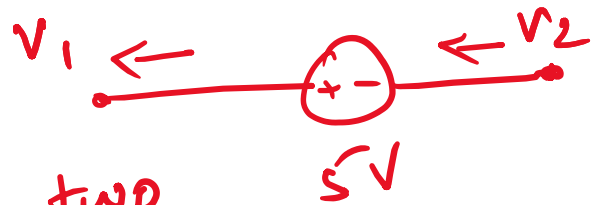
$$x_1 = \frac{\Delta_1}{\Delta}$$

$$x_2 = \frac{\Delta_2}{\Delta}$$

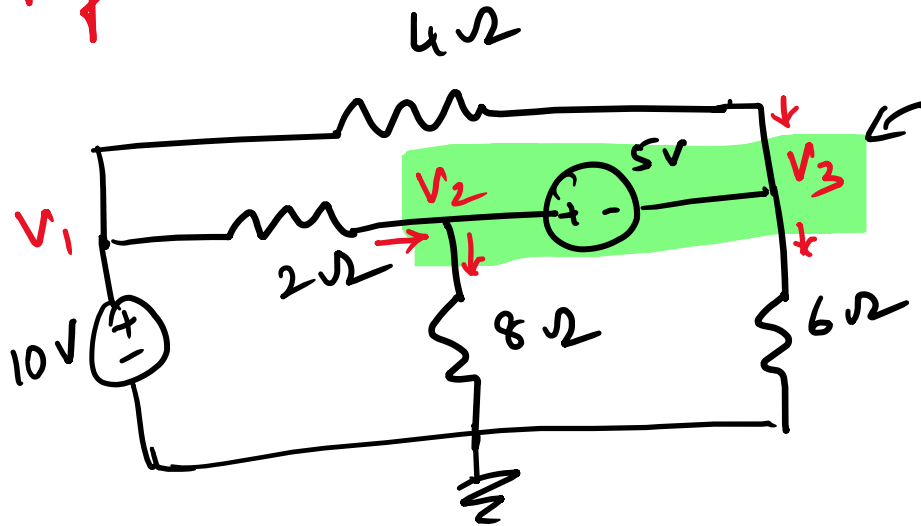
$$x_3 = \frac{\Delta_3}{\Delta}$$

$$\begin{aligned} v_1 &= 0.625V \\ v_2 &= 0.375V \\ v_3 &= 1.625V \end{aligned}$$

Supernode :



Voltage source between two non-reference nodes.



Supernode.

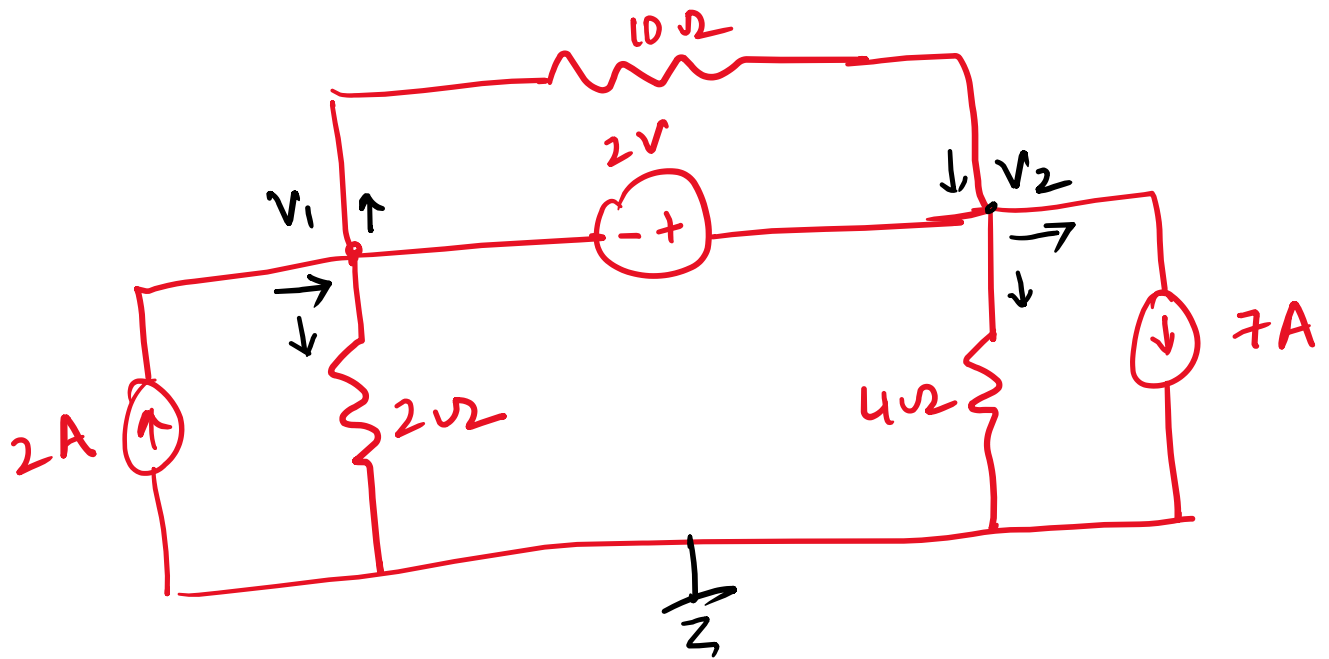
$$V_2 - V_3 = 5V$$

$$V_1 = 10V$$

Supernode : Total incoming into supernode
= total outgoing from supernode

$$\frac{V_1 - V_2}{2} + \frac{V_1 - V_3}{4} = \frac{V_2}{8} + \frac{V_3}{6}$$

$$\begin{aligned} V_2 &= 9.2V \\ V_3 &= 4.2V \end{aligned}$$

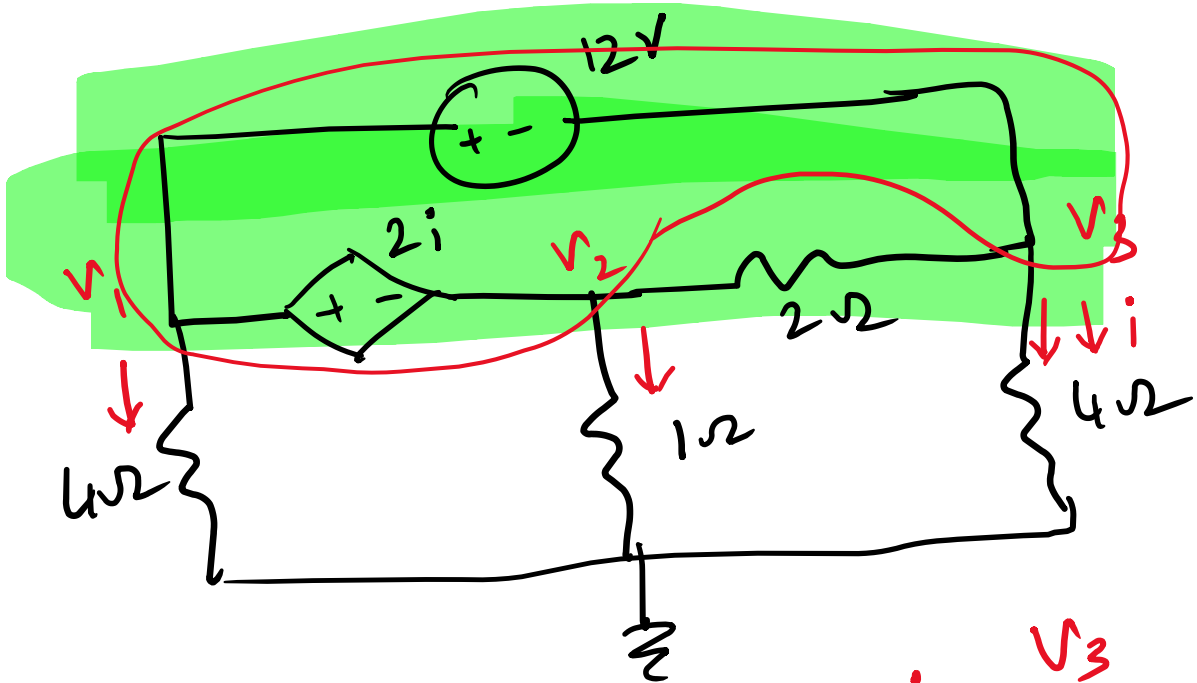


$$v_2 - v_1 = 2$$

$$2 + \frac{v_1 - v_2}{10} = \frac{v_1}{2} + \frac{v_2}{4} + 7 + \frac{v_1 - v_2}{10}$$

$$v_1 = -7.33V$$

$$v_2 = -5.33V$$



$$V_1 - V_3 = 12V$$

$$i = \frac{V_3}{4}$$

$$V_1 - V_2 = V_3/2$$

$$\frac{V_1}{4} + \frac{V_2}{1} + \frac{V_3}{4} = 0$$

$$\begin{aligned} V_1 &= -3V \\ V_2 &= 4.5V \\ V_3 &= -1.5V \end{aligned}$$